Fiscal Year:	FY 2019	**Task Last Updated:**	FY 03/03/2019
PI Name:	Chang, Polly Y. Ph.D.		
Project Title:	Particle Radiation-Induced Dysregulation of Protein Homeostasis in the Brain		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	SRI International		
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City:	Menlo Park	**State:**	CA
Zip Code:	94025-3453	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
Start Date:	05/01/2014	**End Date:**	02/29/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Extended to 2/29/2020 per NSSC information (Ed., 10/16/19) NOTE: Extended to 10/31/2019 per NSSC information (Ed., 5/10/19) NOTE: Extended to 4/30/2019 per NSSC information (Ed., 4/11/18) NOTE: Change in period of performance (original period of performance was 2/26/2014-2/25/2018) per PI and NSSC information (Ed., 12/9/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lin, Hua Ph.D. (SRI International) Shaler, Thomas Ph.D. (SRI International)		
Grant/Contract No.:	NNX14AE69G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The space atmosphere is comprised of wide spectrum of highly energetic and charged particle radiation. Exposure to this type of radiation in the galacto-cosmic environment or unplanned solar particle events (SPE) may pose a potential health hazard to humans in long-term manned space explorations. In normal biological systems, protein stability and system homeostasis are maintained through a tightly coupled post-translational modification process that is catalyzed by the ubiquitin-proteasome system for protein turnover. This process is necessary for modifying the correct protein in the correct cellular location at the correct time to control important processes, e.g., cell-cycle regulation, the maintenance of genomic stability, removal of damaged or misfolded proteins, autophagy, apoptosis, and gene transcription. Depending on the radiation dose and beam quality, particle radiation may cause significant damage to proteins and oxidative stress in the cells in the central nervous system, leading to dysfunction of the cellular ubiquitin-related pathways, an increase in the levels of extracellular ubiquitin, and persistent subsequent downstream effects on regulation of stress-associated signaling pathways in many different cell types within the central nervous system (CNS), precipitating neurodegenerative disease. We hypothesize that quantitative changes in cellular ubiquitin pools occur when neuronal cells are exposed to particle radiation and aim to study the impact of protein degradation after particle radiation using both in vitro and in vivo model systems. Information gathered from this research will aid in reducing the uncertainties associated with predicting risks both in the estimation of risk to astronaut health and the design of appropriate countermeasures to ameliorate any potential negative effects in space flights.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We expect that results associated with the regulation of protein processing and degradation in brain tissues obtained from our studies will be useful to understand molecular mechanisms of action in normal neuronal cell responses in particle radiation clinical therapy scenarios.
Task Progress:	The objective of this project is to conduct in-depth investigations of pathways that regulate the cellular levels of proteins through protein modifications and the role of these pathways in neurodegeneration as a consequence of the exposure. We hypothesize that when cells of important organ systems are exposed to particle radiation, abnormal and persistent changes in cellular ubiquitin pools occur both in the short term, as part of damage repair processes, in response to oxidative stress mediated by the Nrf2 pathway, and in the long term when irreparable damage is present in tissues of very low cellular turnover rates such as CNS. These changes can accumulate as a result of the initial irradiation event, leading to an increase in the amount of polyubiquitin (pUb) chain-modified proteins. Results from our work showed particle radiation quality plays an important role in the level, linkage, and spatial specificity of protein ubiquitination in neuronal cells and brain tissues. We demonstrated differential pUb linkages in neuronal cells depending of the LET (linear energy transfer) of the particle beam. At low LET, increased K48 pUb linkage is seen across the different human and murine cell types, while at high LET, increased K63 repair-associated linkages are seen in the same cells. On the other hand, polyubiquitination in the hippocampus region of C57Bl6 mice was not significantly altered when compared to the sham-treated controls for particles beams with LET ~70 keV/µm. On the other hand, K48 linkages are elevated in a dose-dependent manner as early as 1 wk post irradiation for 1 GeV/u iron ions. The relevance of our current work is supported by our early findings of protein ubiquitination in brain tissues in Alzheimer (AD) patients. The significance of brain ubiquitylome in degenerative brain pathology is further verified by a recent paper by Abreha et al. (Proteomic 18, 2018) characterizing ubiquitin-positive pathological protein aggregates in 3 AD and 5 age-matched control subjects. Detailed analysis of hyperphosphorylation of microtubule associated protein Tau, core component of neurofibrillary tangles, showed that specific sites of Tau protein in specific brain regions (cerebellum and hippocampus) were hyperphosphorylated in brain tissues at 16 months after a low dose of iron ions. This significant finding points to the fact that particle radiation impacts the cross talk of ubiquitation and phosphorylation of Tau proteins and may have a long-term effect in the CNS. Reference Abreha MH, Dammer, EB, Ping, L, Zhang, T., Duong, DM, Geraing, M, Lah J, Levey, A, Swyfried,T. Quantitative Analysis of the Brain Ubiquitylome in Alzheimer’s Disease. Proteomic, 18(20), 2018.
Bibliography Type:	Description: (Last Updated: 04/26/2023)
Abstracts for Journals and Proceedings	Chang PY, Shaler TA, Lin H. "Particle-radiation induced dysregulation of protein homeostasis in the brain." 64th Annual Meeting of the Radiation Research Society, Chicago, IL, September 23-26, 2018. 64th Annual Meeting of the Radiation Research Society, Chicago, IL, September 23-26, 2018. , Sep-2018
Abstracts for Journals and Proceedings	Chang PY, Shaler TA, Lin H. "Investigation into the effects of Particle radiation on the ubiquitin proteasome system in neuronal stem cell and brain tissues." Presented at the 2019 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 22-25, 2019. 2019 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 22-25, 2019. , Jan-2019
Abstracts for Journals and Proceedings	Goukassian D, Cimini M, Garikipati S, Addya S, Blakely E, Chang P, Kishore R. "Cardiovascular disease risks due to single full body gamma and HZE radiation—Long term dose response studies." Presented at the 2019 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 22-25, 2019. 2019 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 22-25, 2019. , Jan-2019